GRADE 10 EXAM BREAKDOWN 2026

PAPER 1	Section	Marks	Total
2 hrs			± 135
	Number Systems(Term 2 booklet) Types of numbers N, N_0, Z, Q, Q', R, R' Recurring → Rational Number Lines Interval notation	± 5	
	Number Patterns (Term 2 booklet) Complete patterns Find formulae. *1st diff formula given* 2nd diff formula not given	± 15	
	Algebra- (Term 1 booklet) Products Factors, NB cubes Fractions	± 15	
	Exponents (Term 2 booklet) $x^a \times x^b = x^{a+b}$ $x^a \div x^b = x^{a-b}$ $(x^a)^b = x^{ab}$ $x^{-m} = \frac{1}{x^m}$ $x^0 = 1$ $x^{\frac{m}{n}} = \sqrt[n]{x^m}$	± 15	
	Equations (Term 1 booklet) Linear Fractional Quadratic Literal Exponential Simultaneous Word Problems Inequalities **(Term 2 booklet)**	± 25	
	Functions (Term 2 booklet) Straight Line Parabola Hyperbola Exponential	± 35	
	Finance (Term 4 booklet) Simple – *Formula given* Compound- *Formula given* Adjusted compound interest – *Formula not given* Time Lines	± 10	
	Probability (Term 4 booklet) Tree diagrams Venn Diagrams Basic theory *Formulae below are given* $P(A) = \frac{n(A)}{n(S)}$ *P(A or B) = P(A) + P(B)-P(A and B)*	± 20	

PAPER 2	Section	Marks	Total
2 hrs			± 120
	Trigonometry (Trig booklet T3) Sinθ, cosθ,tanθ definitions Calculator work Finding missing sides and angles using trig ratios CAST-all quadrants Pythagoras type problems Trig equations Trig graphs- interpretation; effect of a and q; transformation of graphs	± 35	
	Data Handling (Term 4 booklet) Histograms, frequency polygons, class intervals, midpoints, mean, median, mode, range, quartiles, IQR, percentiles, deciles, box-and-whisker plots, skewness.	± 20	
	Analytical Geometry (Term 3&4 booklet) *Formulae are given* Distance formula, midpoint, gradient, inclination, collinear, 4th point of a parallelogram, quads	± 30	
	Measurement (Term 2 booklet) Surface area, Volume, Scale Factor	± 5	
	Euclidean Geometry (Geometry booklet T1) Basic grade 8 Geometry Properties of quadrilaterals Similarity/ congruency Prop Intercept theorem	± 35	

Further revision options:

MAS end of chapter revision exercises and challenges

Visit the following sites and select only the worksheets /activities that are based on what you have studied this year:

https://www.siyavula.com/read/za/mathematics/grade-10

https://www.mathsatsharp.co.za/list-of-all-worksheets

https://www.kutasoftware.com/freeige.html

GRADE 10 REVISION NOVEMBER

PAPER 1

A Numbers & Patterns

QUESTION 1
Classify the numbers by placing the ticks in the appropriate columns

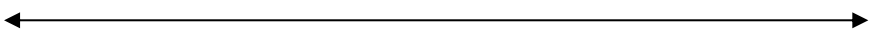
Number	R	Q	Q'	Z	N_o	N	Non-real
$-4\frac{1}{2}$							
0,1458							
$\sqrt{16}$							
$0,\dot{4}$							
$\sqrt{39}$							
-5							
$\sqrt{-4}$							
π							
9,23647…..							

QUESTION 2
Answer True or False. Tick the correct block.

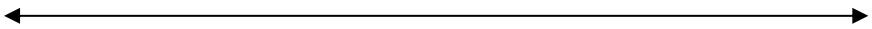
2.1	Integers include all the natural numbers.	T	F
2.2	The number 1,25 is a whole number.	T	F
2.3	The number $-1\frac{1}{2}$ is an integer	T	F
2.4	Rational numbers include all the integers and all the fractions	T	F
2.5	$\sqrt{7}$ lies between the numbers 3 and 4	T	F

QUESTION 3
Represent the following sets on a number line.

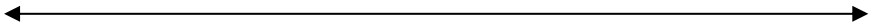
3.1 $\{ x \ / \ x \geq 4 \ ; \ x \in R \}$



3.2 $\{ x / -2 < x \leq 3 ; x \in N \}$



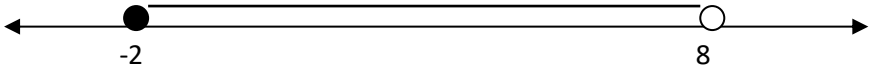
3.3 $x \in (2 ; \infty)$



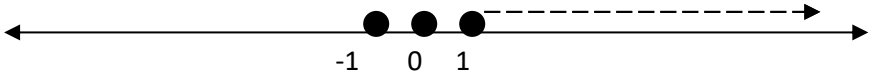
QUESTION 4

Write the following in set builder or interval notation. Give each answer under each number line.

4.1



4.2



QUESTION 5

Convert $0,\dot{7}$ to a fraction. Show your working.

QUESTION 6

6.1 Complete each pattern by writing in the next two terms each time:

6.1.1 $m+1;\ \ m-1;\ \ m-3;$

6.1.2 $1;\ \frac{1}{8};\ \frac{1}{27};$

6.2.1 $T_n = 2n+1$

6.2.2 $T_n = 2n^2$

6.2.3 $T_n = 3.2^n$

QUESTION 7

Simplify the following, give your answers with positive exponents:

7.1 $2a^2 \times 3a^3$

7.2 $\dfrac{16a^{\frac{3}{4}}}{8a^{\frac{1}{4}}}$

7.3 $0{,}5^0 \times 0{,}5^{-2} \times 9^{\frac{1}{2}}$

QUESTION 8

A two digit number is exactly divisible by 12 ; 14 and 21. Find this two digit number.

QUESTION 9

Look at the following number pattern

$1 + 3 = 4 = 2 \times 2$

$1 + 3 + 5 = 9 = 3 \times 3$

$1 + 3 + 5 + 7 = 16 = 4 \times 4$

9.1 What is the sum of the first 10 odd numbers?

9.2 What is the sum of the first n odd numbers?

QUESTION 10

A petrol tank holds 71*litres*. The tank already contains 5*litres.* A petrol pump fills the tank at 3*litres* per 10 seconds.

a. How long will it take the tank to fill to 23 litres, to the nearest minute?

b. How long will it take to fill the tank, to the nearest minute?

B. PRODUCTS , FACTORS, FRACTIONS,EQUATIONS

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1. Find the following products. Simplify your answer fully:

a) $(2x-5)(3x + 5)$

b) $(2x – 3y)^2$

c) $(p - 3)(p^2 + 3p + 9)$

d) $\left(x - \sqrt{7}\right)\left(x + \sqrt{7}\right)$]

2. Expand the following and simplify fully:

a) $x(x + y) – (x + y)^2 - 2$

b) $[- 4(2p + q)]^2$

c) $\left(3x - \frac{2}{x}\right)\left(x + \frac{1}{x}\right)$

d) $(a^b + 1)(a^b – 1)$

3. Factorise completely:

a) $121 - p^2$

b) $a^2 + 3a + 2$

c) $2x^2 – 13x +6$

d) $10p^4 – 10$

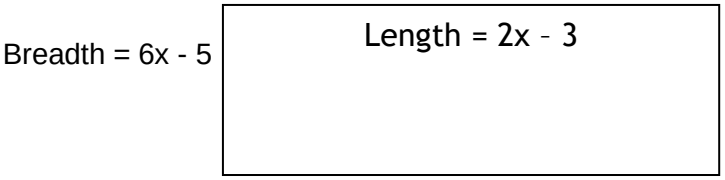
e) $a(x – y) – b(x – y)$

f) $4x^3(x – 1) + x (1 – x)$

g) $ax – 2bx – 4by + 2ay$

h) $(x - 2)^2 – b^2$

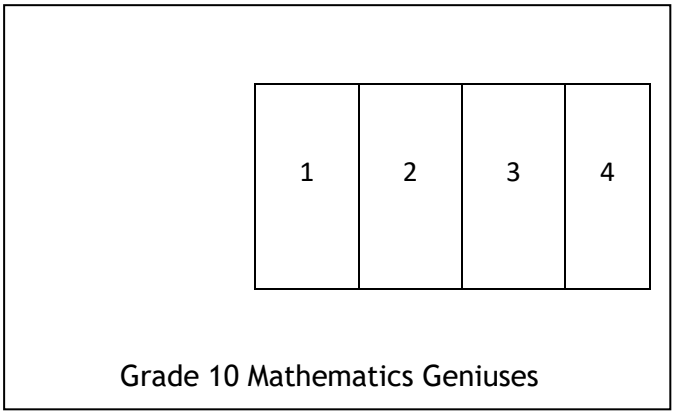
4. Study the rectangle below:



a) Give an expression for the area of the rectangle.

b) If each side of the rectangle is doubled in length, how will the area increase? Explain or show working.

5. In how many different combinations can four different stamps be stuck **in a straight row,** as shown, on an envelope? Show all working!



Fractions:

Simplify each of the following:

1.1. $\frac{12a^2b^3}{20ab^4}$

1.2. $\frac{9x^2-y^2}{4} \div \frac{9x+3y}{2}$

1.3. $\frac{2b}{a^2-b^2} + \frac{b}{ab-a^2}$

Question 3:

Solve for x and/or y:

3.1 $x(x+2) = (x+1)(x+4)$

3.2 $\frac{3}{5x} - \frac{2(x+3)}{3x} = 1 + \frac{2}{x}$

3.3. $3y^2 - 12y = 15$

3.4. Kendall will be four times older than she was ten years ago in five years time.

How old is she now?

Question 4:

4.1. Are the following statements true:

$$2^2 - 1^2 = 2 + 1$$

$$4^2 - 3^2 = 4 + 3$$

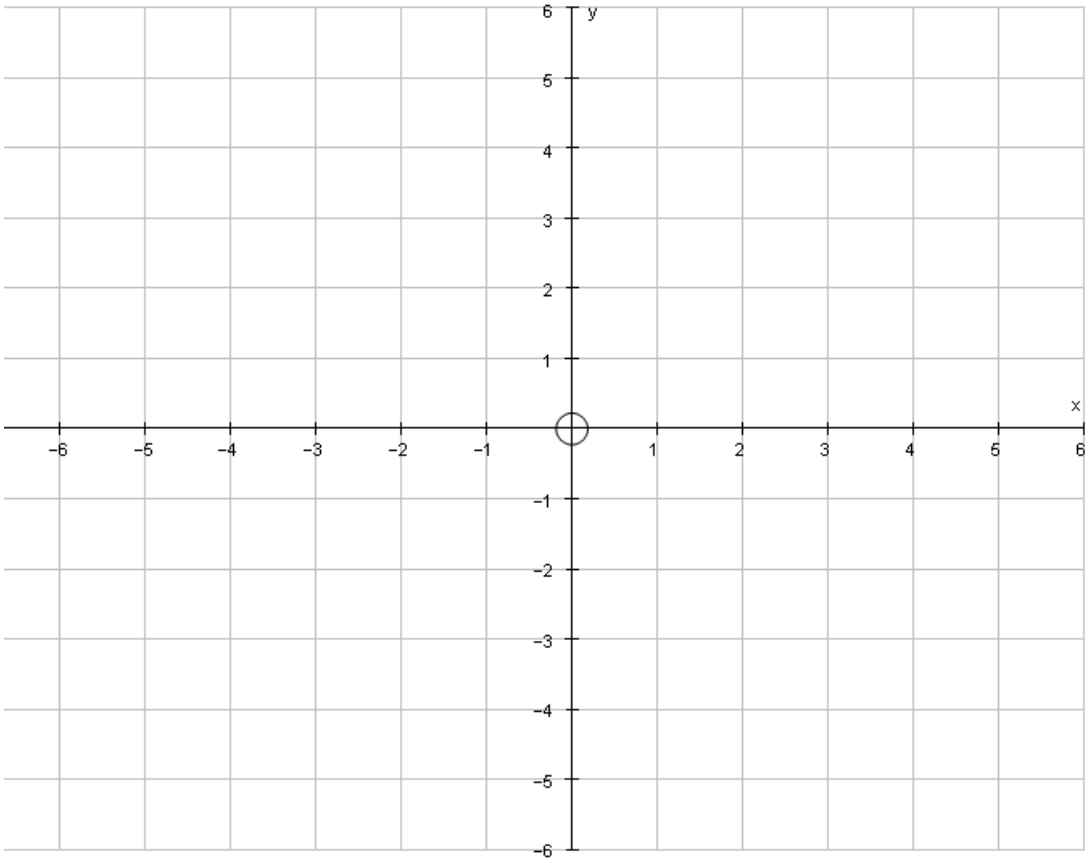
4.2. Continue the above pattern by writing the next line

4.3. Prove algebraically that the above pattern is true for any *n.*

C. FUNCTIONS

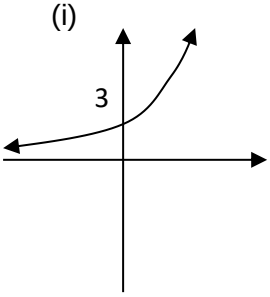
1. Neatly draw all the given graphs on the axes below. Label all intercepts.

$f(x) = -x^2 + 4$ $\quad$ $g(x) = 2$ $\quad$ $h(x) = (\frac{1}{2})^x$

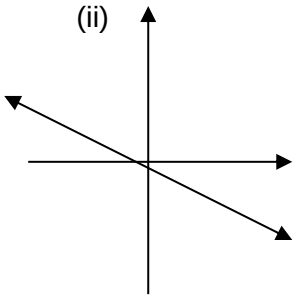


2a)Match each graph with **one** equation:

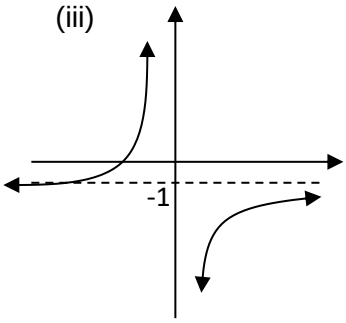
(i)



(ii)



(iii)



1. $y = x$ $\quad$ **2.** $y = -\frac{1}{2}x$

3. $y = 2^x$ $\quad$ **4.** $y = 2^x + 2$

5 $\quad y = x^2 + 1$ $\quad$ **6.** $y = \frac{4}{x} - 1$

7. $y = -\frac{4}{x} - 1$

(b) For the graph iii above, give the following:
(A) domain
(B) range
(C) asymptotes (if possible)

D. EXPONENTS

3. Solve for *m*:

$5^{m-1} = 25$

4. Simplify:

(a) $(-3m^0).3^{-1}$

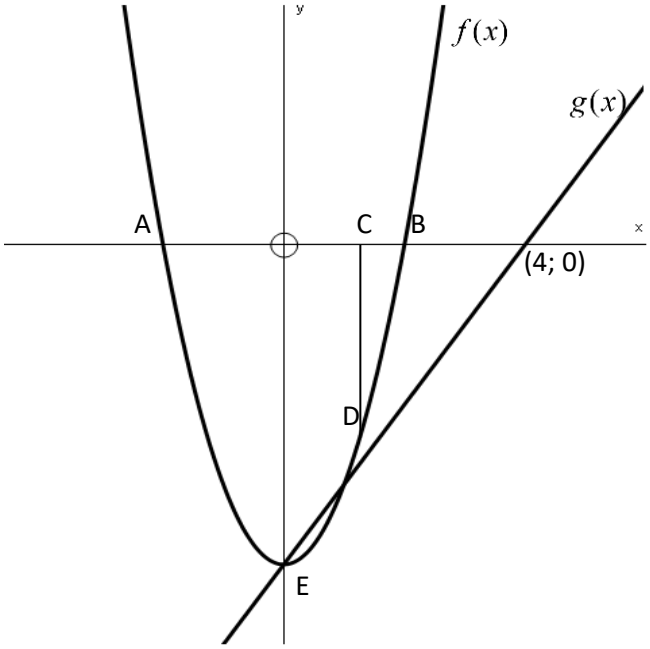
(b) $\dfrac{-6xy^3 \times 6x^5y^4}{12x^6y^8}$

(c) $\dfrac{32^x.2^{-2x-2}}{8^{x-2}}$

(d) $\sqrt[4]{x^5}.(x^3y^2)^{\frac{1}{4}}.x^{-2}$

E. MORE FUNCTIONS

5. The sketch represents the graph of $f(x) = 2x^2 - 8$



(a) Calculate the x-co-ordinates (A and B) and the y-co-ordinate (E) of $f(x)$.

(b) A line $g(x)$ goes through E and (4; 0). Find its equation.

(c) What is the length of AB?

(d) Calculate f(2)

(e) Calculate f(a + 1)

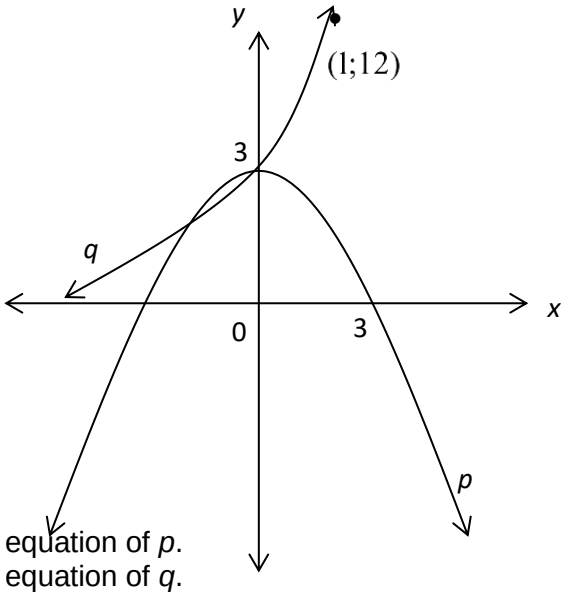
(f) Calculate p if f(p) = -6

(g) Is $f(x)$ a function? Give a reason.

(h) Write down the axis of symmetry for $f(x)$.

(i) If OC is 1 unit, find the length of CD.

6. Functions $p: y = ax^2 + q$ and $q: y = a.b^x$ intersect at $(0;3)$. The parabola has an *x*-intercept of 3 and the point (1;12) lies on *q*.



(a) Determine the equation of *p*.
(b) Determine the equation of *q*.
(c) For which values of *x* are both graphs increasing?

7. If $f(1) = 5$ and $f(x+1) = 2f(x)$, what is the value of $f(7)$?

F. FINANCE

1. R6 578 is invested for 48 months at 9,5%p.a. simple interest .

How much is in the account at the end of the 48 months?

2. A trolley of groceries cost R250 in January 2003.

a) Calculate the expected cost of a trolley of the same groceries in January 2010 if the average rate of inflation is 9%p.a.

b) What would the same trolley of groceries have cost in January 2001?

3. Mrs Smith buys a fridge and deep freezer for R8 700. She pays for them over 3 years on a hire purchase agreement. The interest rate is 12%p.a. based on the full purchase price over three years.

a) How much will she end up paying for these items over the 3 years?

b) How much will she pay each month if an insurance fee of R22 is added to each month's payment?

4. An investment triples in value after 5 years if compounded annually.

At what rate was the investment compounded?

5. I deposit R6000 in the bank for my 7 year old daughter's 18th birthday.

   I give her an amount of R20 000 on her birthday (after closing the account).

   What rate did I receive on my investment, if simple interest was calculated?

6. R 5 650 is invested in account for 5 years. Interest is set at 10,5% compounded weekly. How much money will be in the account at the end of the 5 year period?

7. An investment of R12 500 earned 12% p.a. for 4 years compounded annually.

   The interest rate then dropped to 11% p.a. for the next 3 years.

   Calculate the final amount in the account after the 7 years.

G. PROBABILITY

1. A box contains coloured blocks. The number of each colour is given in the following table:

Colour	Purple	Orange	White	Pink
Number of blocks	24	32	41	19

A block is selected randomly. What is the probability that the block will be:

a) Purple?
b) Purple or White?
c) Pink and Orange?
d) Not Orange?

2. In a survey 70 people were questioned about which product they use: A or B or both. The results show that 25 people use product A, 36 people use product B and 15 use neither.
   a) Draw a Venn diagram to represent this information

   b) How many people in the survey use product B only?

   c) If a person is selected randomly from the sample group, what is the probability that they use sample B only?

   d) Is the use of the products mutually exclusive? Give a reason for your answer.

   e) If a random selection is made from the survey group, what is the probability that the person selected uses product A and B?

3. Two events, P and Q, are complementary. The probability of P occurring is $\frac{2}{5}$.
   a) What is the probability of Q occurring?
   b) What is the probability of Q not occurring?

4. a) A person is selected randomly from your class. What is the probability that they have a birthday this year on a Monday?

   b) Draw a probability tree to illustrate this, use M and M'

   c) A second person is chosen randomly from your class. Continue your tree diagram to allow for this person's birthday being on a Monday or not this year.

   d) What is the probability that both people will have birthdays on a Monday this year?

   e) What is the probability that neither person will have their birthday on a Monday this year?

5. In a card game, all the hearts and all the clubs are removed from the pack.
   A person is asked to select a card from the remaining cards , keep it, and then make a second selection from the remaining cards.

   a) Draw a tree diagram showing the probabilities on both selections.

   b) Using your diagram, give the probability of selecting a diamond both times.

   c) What is the probability of selecting a diamond both times or a spade both times?

PAPER 2

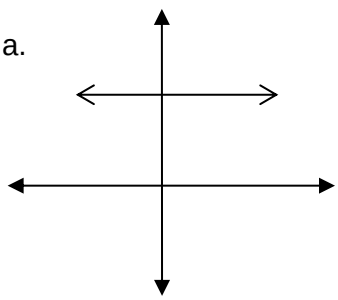
$m = \frac{y_2 - y_1}{x_2 - x_1}$ $\quad$ $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$M = \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2} \right)$ $\quad$ $y = mx + c$ $\quad$ $\tan\theta = m$

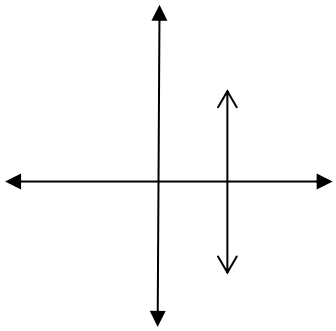
A. Analytical Geometry

Question 1
Determine the gradient of the following lines:

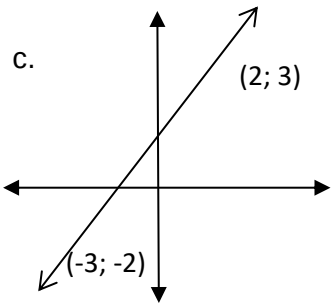
a.



b.



c.



Question 2. Determine the angle of inclination of each line in Question 1.

Question 3

a. Determine the midpoint of the line joining the points (1; 4) and (3; 6).

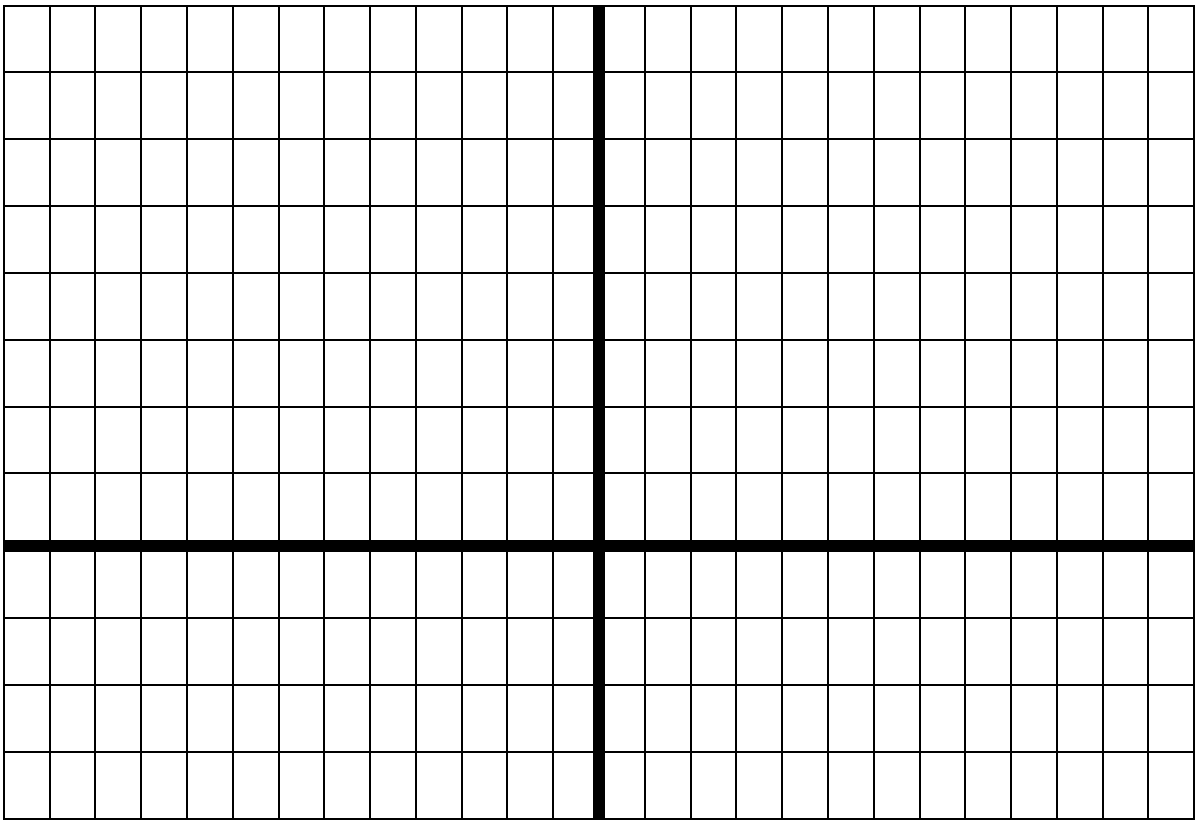
b. Calculate the length of AB if A(– 4; 3) and B (6; 1). Leave your answer in surd form.

c. Find the gradient of the line passing through K(-6; 3) and L (4; 5)

Question 4 Given that P(3; 4), Q(2; 3), R(5; 1) are three vertices of a parallellogram.

a. Plot the vertices on the grid below. Label the vertices.

Plot S so that PQRS is a parallelogram.



b. Calculate the equation of a line parallel to PQ passing through (-1; 6)

e. Will the points Q, R and T(6; -4) be collinear? Show working.

Question 5 A(1; 4); B(5; 4); C(5; 1) and D (1; 1) are points in the Cartesian Plane.

a. Calculate the length of each side of the quadrilateral.

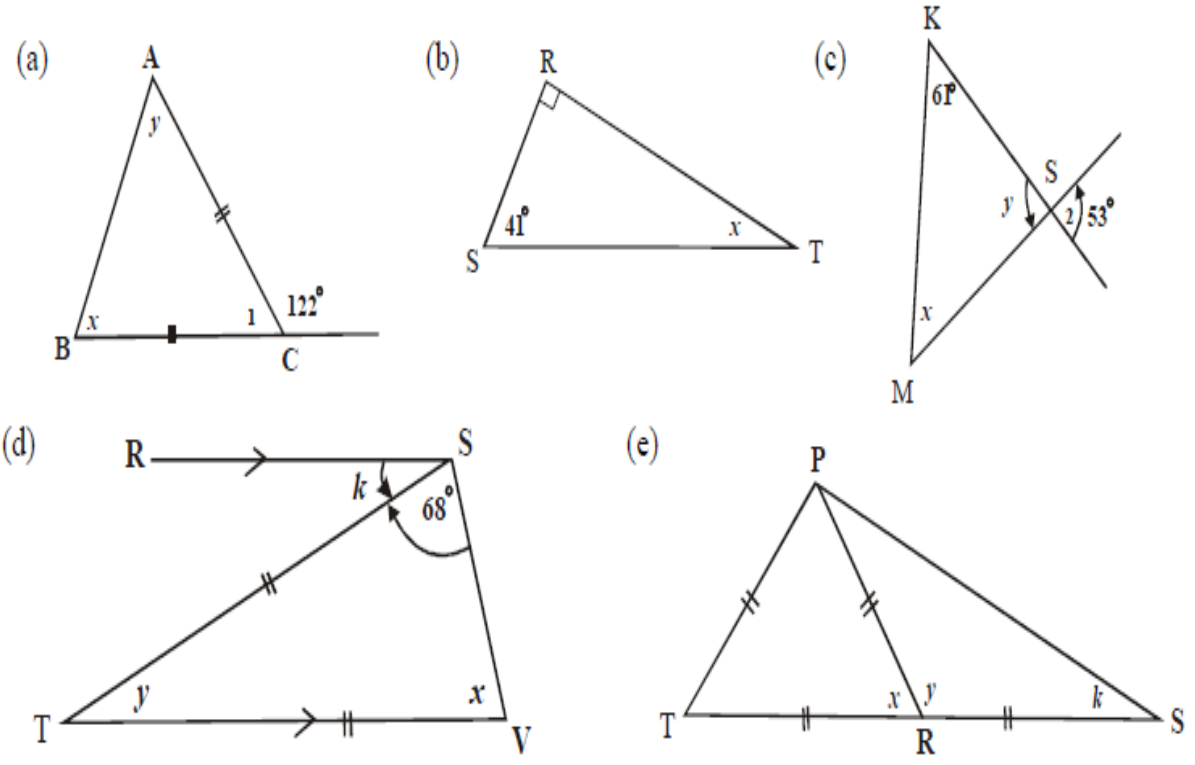
b. What type of quadrilateral is ABCD? Give a reason for your answer.

c. Show that M $\left(3;\frac{5}{2}\right)$ is the point of intersection of the diagonals.

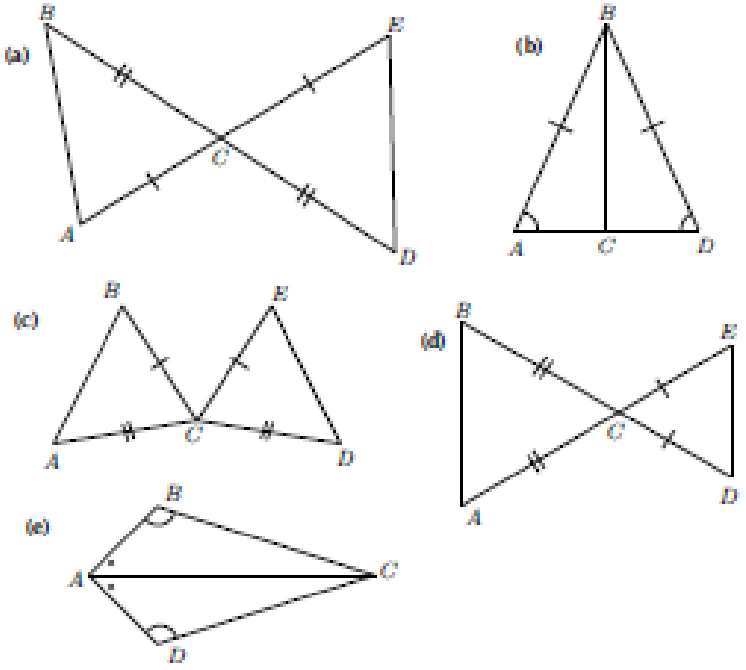
Question 6 Show by calculation, which of the points P(3; 5) or Q(– 4;– 4) is closer to the origin.

B. **EUCLIDEAN GEOMETRY**

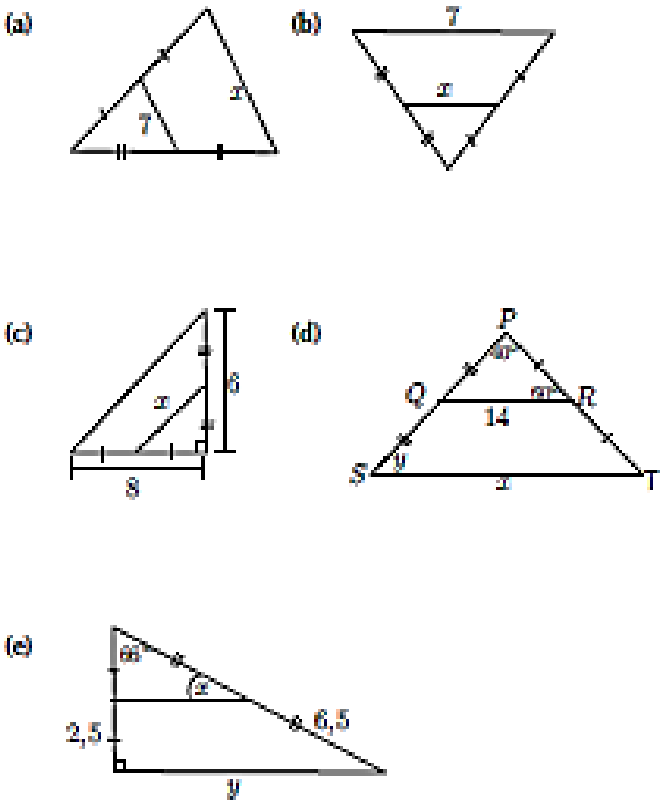
(1) Calculate the values of x, y and k in each of the following and provide reasons:



2. State whether the following pairs of triangles are congruent or not. Give reasons for your answers. If there is not sufficient information to make a decision, say so.

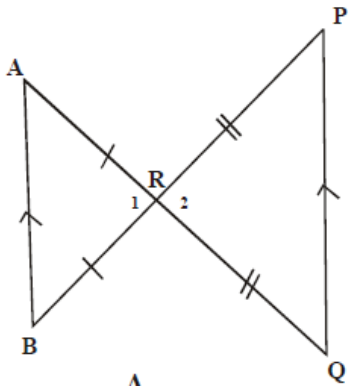


3. Find x and y in each of the following if the required lines are parallel. Give reasons.



4.

AB // PQ, AR = BR and PR = QR.
(a) Prove that: Δ ABR /// Δ QPR
(b) Calculate the length of AB, if PQ = 8 cm, AR = 3 cm and QR = 4 cm.



C. Trigonometry

QUESTION 1

a) If A = 84° and B = 143° , find the value of the following to two decimal places:

i) $5tan\ B =$

ii) $sin\ (B - A) =$

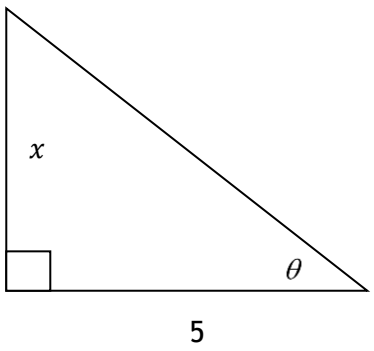
iii) $cos\ \frac{A}{2} =$

iv) $\frac{sin^2 B}{2} =$

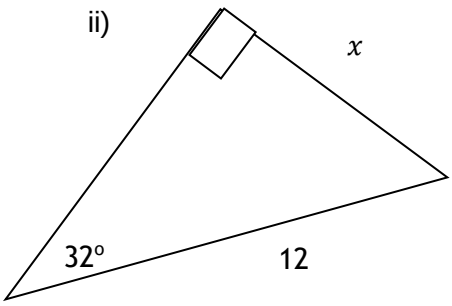
QUESTION 2

a) Find x in each case:

i)



ii)



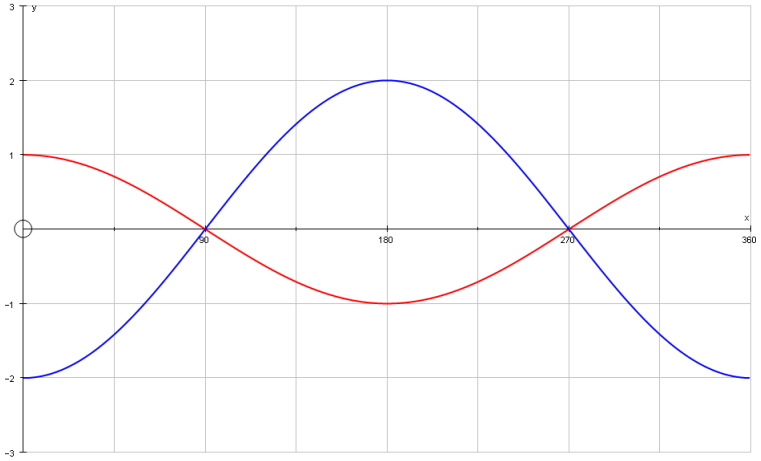
Complete

iii) $\sin\theta =$ iv) $\tan\theta =$ v) $\theta =$

QUESTION 3 *If* $tan\ P = \frac{4}{3}$ *and* $P \in [90°; 270°]$. *Use a diagram and without a calculator determine:* $sin\ P + cos\ P$

QUESTION 4

Below are the sketches of two trigonometric graphs $f(x) = cos\, x$ and $g(x) = -2cos\, x$. Answer the questions that follow:



a) Give the period of $f(x)$

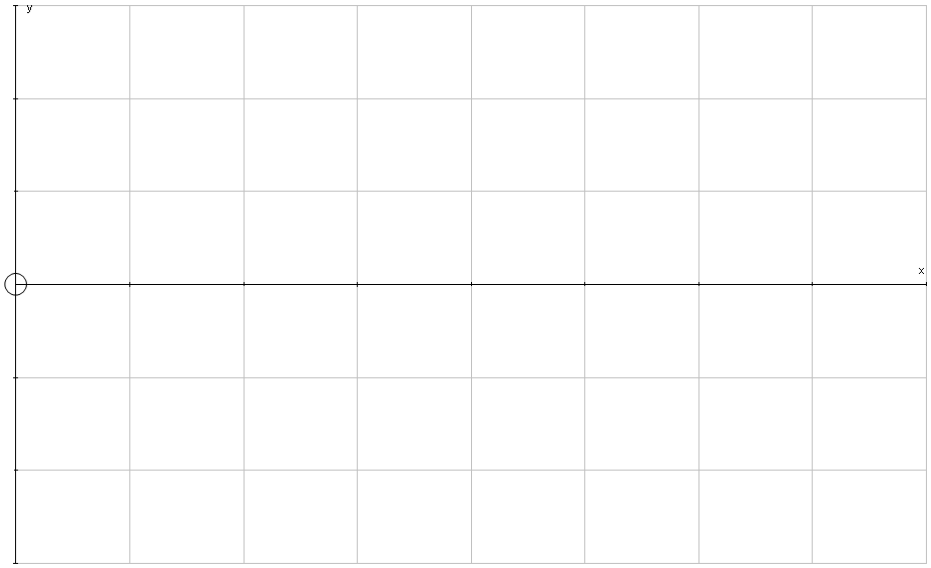
b) Give the amplitude of $g(x)$

c) Give the values of x where $f(x) = g(x)$

d) For which values of x is $g(x) \geq 0$?

QUESTION 5

On the axes provided, sketch the graphs of i) $y = sinx + 1$ **and** ii) $y = 2tan\, x$, $x \in [0°; 360°]$. Indicate your scale on both axes and any asymptotes.



QUESTION 6

Solve for x correct to 1 decimal place:

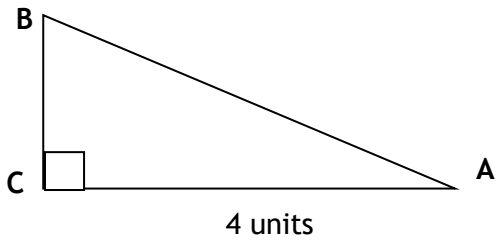
a) $Sin\, x = 0{,}812$

b) $\frac{1}{2} cos x = -0{,}445$

QUESTION 7

In the diagram below ABC is a right angled triangle at C.

Tan A = $\frac{3}{2}$ and A $\in [0^o; 90^o]$, AC = 4 units



Find the following:

a) Cos A =

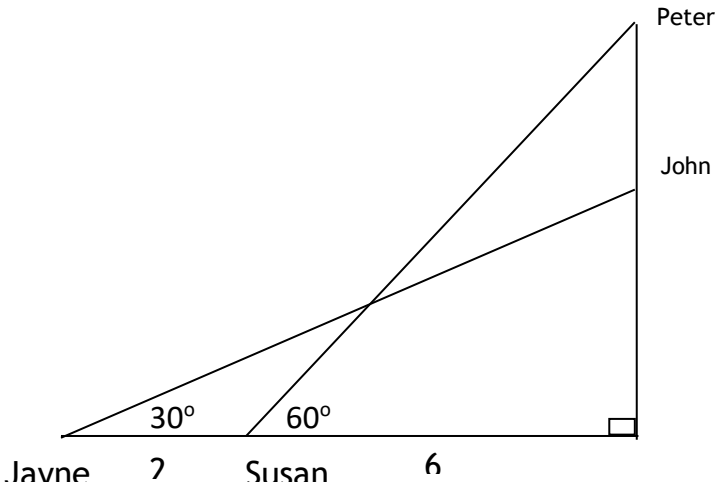
b) length of BC

c) Length of AB correct to 2 decimal places

d) The size of Angle B

QUESTION 8

Susan and Jayne are standing 2 metres away from each other and are talking to Peter and John who are standing at the second floor and first floor windows respectively.



Susan is standing 6 metres away from the building. If the angle of elevation of Jayne to John is 30° and if the angle of elevation of Susan to Peter is 60°. Determine the distance between the two floors giving your answer correct to 2 decimal places.

Data handling

1 Thirteen students wrote a test and their time to complete the test (in minutes) was recorded below:

Student	A	B	C	D	E	F	G	H	I	J	K	L	M
Time	40	30	27	32	27	25	30	29	32	27	25	28	25

(a) What is the range?

(b) Calculate the mean time.

(c) Write down the mode and the median.

(d) Calculate the inter-quartile range.

(e) Draw a box and whisker plot for the given data.

(f) Comment on the skewness of the data.

2. **Over one year the average monthly temperatures (°C) for Cape Town were as follows:**

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
28	29	25	21	18	16	11	14	16	19	22	24

(a) Calculate the following:

(1) the mean temperature

(2) the modal temperature

(3) the median temperature.

(4) the first (lower) and the third (upper) quartile**.**

(5) the range and the inter-quartile range.

(6) Draw a box and whisker diagram to represent this data.

(7) Comment on the skew of the data using a mathematical reason.

3. The heights of 80 grade 12 boys were recorded as follows:

Height (cm)	Number of boys
132 − 140	1
141 − 149	4
150 − 158	10
159 − 167	20
168 − 176	34
177 − 185	7
186 − 194	3
195 − 203	1

(a) Determine (1) the approximate mean
(2) the modal class

4. Comment on why this statement is not true.

DAILY RAG

WINE MAKES YOU TALLER, ITS OFFICIAL!!!!

Yesterday we interviewed 100 women about drinking wine. Fifty said they did and fifty said they didn't. The wine drinkers were on average 2cm taller than the non-wine drinkers.

Positive proof that drinking wine makes you taller!!!